University of Galway
School of Computer Science
CT5209 – Cloud DevOps

CA1: Creating a Continuous Deployment Pipeline for a Web Application

V1.0

Goal:

Develop and Operate a Web Application for Petitions (creating, viewing, and signing petitions) following DevOps automation mechanisms.

Requirements:

1. Web App:

- Name of the Project: <your_first_name>spetitions
    - E.g.,: If you name is John: johnspetitions
- Start with Spring Boot initialiser with Maven, and Java 17
- Create at least:
    - 1 page to create a petition
    - 1 page to view all petitions
    - 1 page to search for a petition + 1 page with the result
    - Each petition could be clicked
        - 1 page to view the petition and sign it (with name and email)
    - It is not mandatory to use a database
        - You could use Java data structures (e.g., lists) to store the data.
        - Put some random data in them at the start.

2. Development

- Use GitHub as the Version Control System
    - Have at least 1 commits per feature (page in the web application)
    - Storing Jenkinsfile in Github
    - Configuring GitHub Webhook
- NOT MANDATORY: Following Test Driven Development
    - For each feature, have at least 2 commits (1 for unit tests and one for code)

3. Continuous Deployment

a. With Jenkins CI/CD pipelines (using Jenkinsfile)
b. Running on Amazon EC2
c. Automatically run after each commit
d. With at least the following stages:
    i. Get the code from Github
    ii. Build
    iii. Test
    iv. Package and archive as a War file
        - named <your_first_name>spetitions.war
          e.g., johnspetition.war
    v. (re-)Deploy, if the user manually approves it

- as a Tomcat container
- On Amazon EC2
- Publicly accessible at http://SOME_EC2_PUBLIC_IP:9090

General Instructions

This CA individual and it is worth **20%.**

What to submit?

You will need to submit a zip file that contains the following:

1. **A PDF Report**

The report you write should be clear, well organized and at a professional standard. This is an indicative structure that you can follow:

- Front Page:
  - title of your report
  - Your name, student ID number, email address,
  - module code
  - Link to your Github repository
    - Should be a public repository.
  - Link to your Web Application on Amazon EC2
    - Should be accessible from anywhere.
- Description and organization of your web application.
- Reflection on the challenges
  - What was hardest in the project?
  - List of requirements that you have not (fully) covered.
    - How you might try to cover them if given more time
- Any appendix

The total number of pagers for this report is a maximum of 5 pages (excluding the appendix, the Front page). This is a maximum and not a target.

2. **A 5 Mins (Max) video demo:** showing and explaining the working of your implementation/pipeline.

- You need to particularly show the full process in action:
  i. The Web application before change
  ii. change, commit, automatic build,
  iii. asking the developer if they want to deploy,
  iv. The Web application after change

Marking scheme:

Item	**Mark (/100)**	**Fail (<40%)**	**Pass (40%--59%)**	**Excellent (>60%)**
Req 1	20%	No web application or only the Spring Boot initializer	Web application implementing most of the features	Web application with all the described features
Req 2	20%	No use of Github.	Use of Github but very few commits.	Using Github with enough commits to show the iterative development. Configuration of Webhook.
Req 3	40%	No pipeline or very limited pipeline	Pipeline with most stages	Pipeline with all stages. Automatically run after commit. Automatic Deployment after confirmation.
Report and Code Quality	10%	No report is provided, the report is poorly written, or the report does not follow the required structure. Links to Github/EC2 not provided.	The report is provided, well structured, covers all the required sections but contains some inaccuracies, spelling/grammatical errors typos. Link to code provide, but code/tests are not well documented. Not reflection on challenges and non-implemented features.	The report is at a professional level: well structured, has no inaccuracies, errors. Link to Github provided and code/tests well documented. Good reflection on challenges
Video Demo	10%	No video demo or video of a very poor quality is provided. Video not describing the implementation and the working of the application.	Video demo is provided, but the description of the implementation and working of the application and CI/CD are not detailed/clear.	Video demo is provided. The video demo provides a clear overview of the implementation and showcases the functioning of the application and CI/CD pipeline.